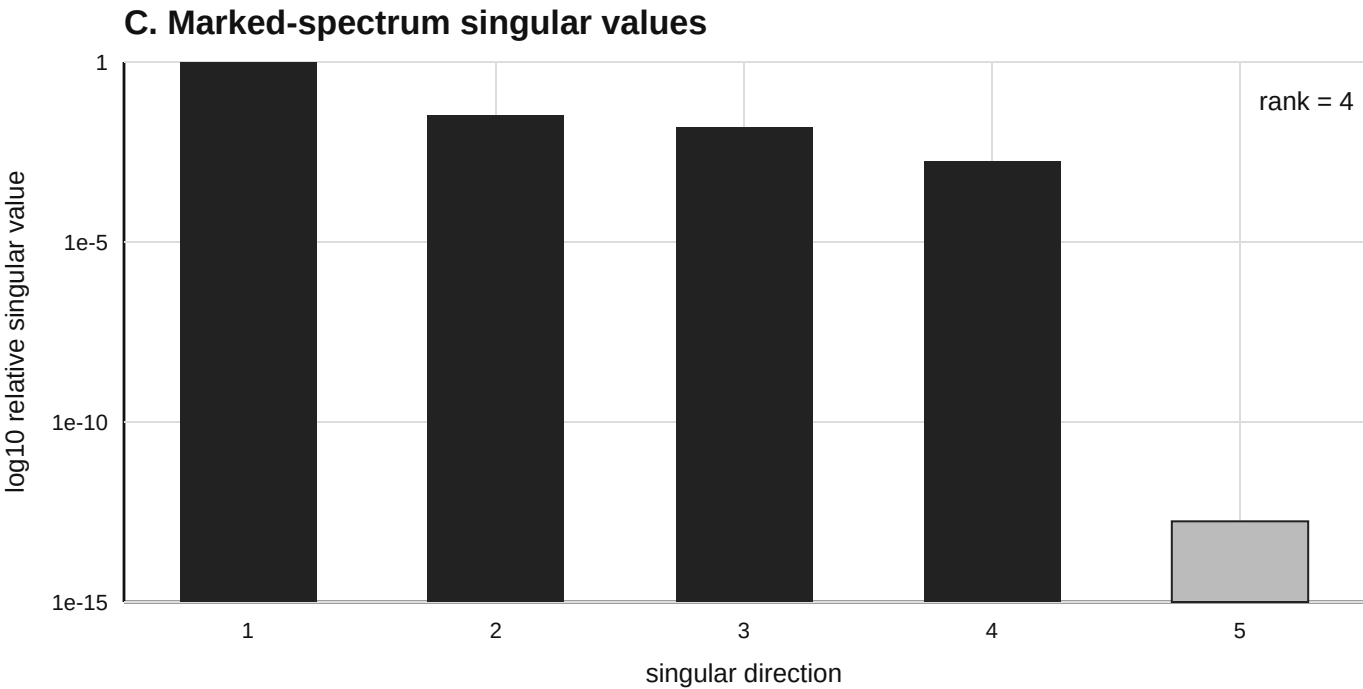
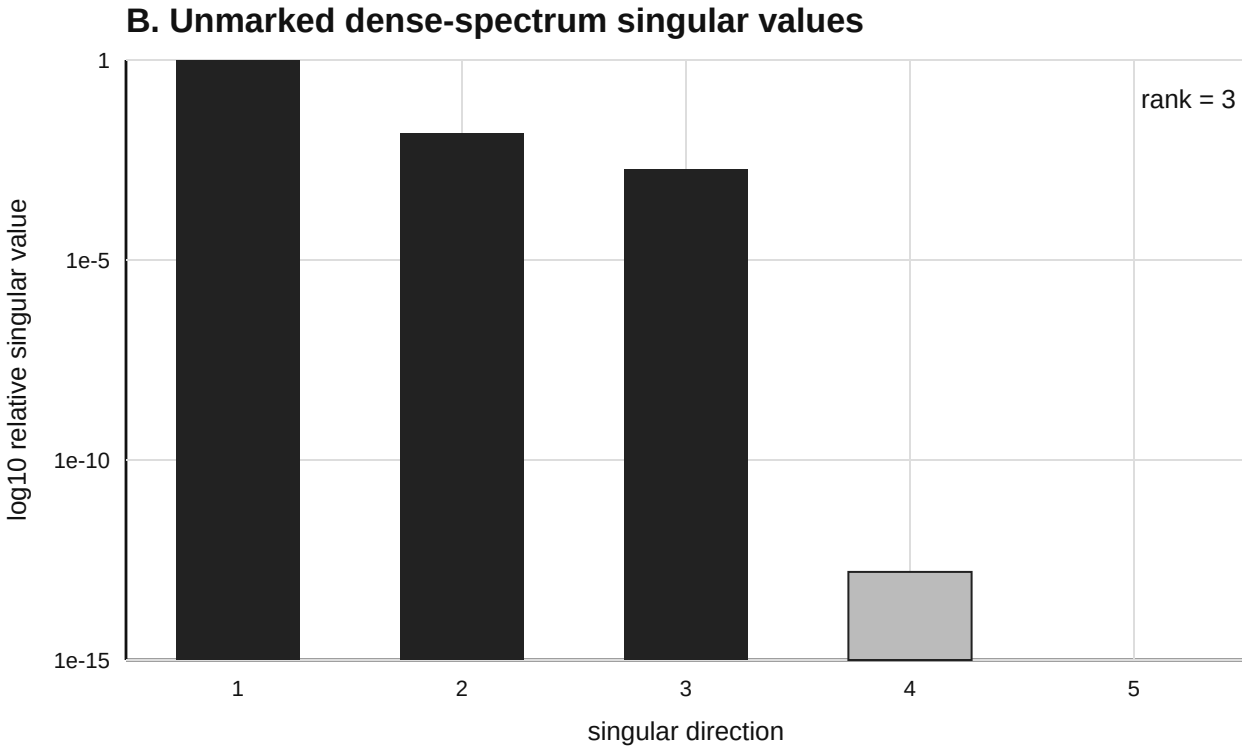
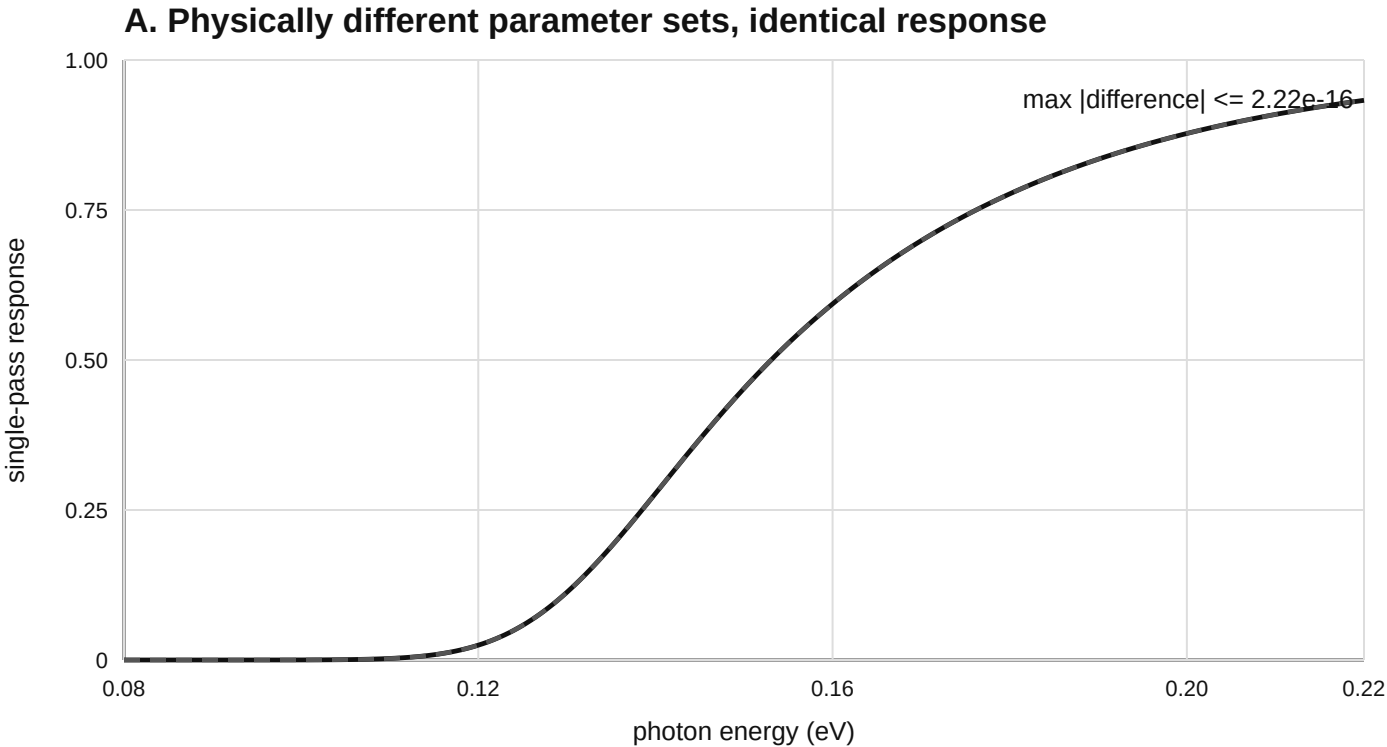


Figure 6. Exact spectral equivalence and structural rank

EXACT FORWARD-MODEL THEOREM WITH SYNTHETIC COUNTEREXAMPLE - NOT MATERIAL VALIDATION



D. Exact invariant combinations

Unmarked: $Eg_0 + \Delta$, σ_G , $A \cdot d$

Marked adds: $\Delta \cdot d$

remaining null vector:

(Δ , $-\Delta$, 0, -1 , $+1$)

Known effective thickness removes the marked-model null.